

GED Math: Ratios, Proportions, and Percents

November 6, 2025

Lesson 1: Understanding Ratios

Objective: Understand how to express and simplify ratios.

- A **ratio** compares two quantities by division.
- Ratios can be written as $a : b$, a/b , or "a to b".
- Simplify a ratio the same way you simplify a fraction.

2 parts blue 2 parts red



Worked Examples: Ratios

Example 1: Simplify $12 : 16$.

Solution: $\frac{12}{16} = \frac{3}{4}$, so the ratio is $3 : 4$.

Example 2: Express 8 red marbles to 12 blue marbles as a ratio.

Solution: $8 : 12 = 2 : 3$.

Example 3: Write the ratio of 50 cm to 2 m.

Solution: Convert $2 \text{ m} = 200 \text{ cm}$. Ratio $= 50 : 200 = 1 : 4$.

Example 4: If a class has 10 boys and 15 girls, ratio of boys to total?

$10 : (10 + 15) = 10 : 25 = 2 : 5$.

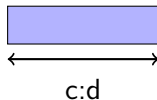
Example 5: A recipe calls for 4 cups of flour and 2 cups of sugar. Simplify.

$4 : 2 = 2 : 1$.

Lesson 2: Understanding Proportions

Objective: Solve problems involving proportions.

- A **proportion** states that two ratios are equal: $\frac{a}{b} = \frac{c}{d}$.
- Use **cross multiplication**: $a \times d = b \times c$.



Worked Examples: Proportions

Example 1: $\frac{3}{4} = \frac{x}{12}$

$$3 \times 12 = 4x \Rightarrow 36 = 4x \Rightarrow x = 9$$

Example 2: $\frac{8}{x} = \frac{2}{3}$

$$8 \times 3 = 2x \Rightarrow 24 = 2x \Rightarrow x = 12$$

Example 3: If 5 pencils cost \$2.50, how much do 8 pencils cost?

$$\frac{5}{2.50} = \frac{8}{x} \Rightarrow 5x = 20 \Rightarrow x = 4.00$$

Example 4: $\frac{7}{14} = \frac{x}{28}$

$$7 \times 28 = 14x \Rightarrow 196 = 14x \Rightarrow x = 14$$

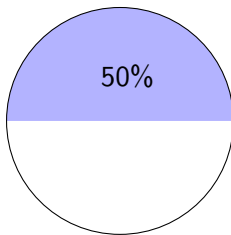
Example 5: If 3 gallons of paint cover 900 sq ft, how many sq ft for 5 gallons?

$$\frac{3}{900} = \frac{5}{x} \Rightarrow 3x = 4500 \Rightarrow x = 1500 \text{ sq ft.}$$

Lesson 3: Understanding Percents

Objective: Solve problems involving percent, part, and whole.

- Percent means **per hundred**. $50\% = \frac{50}{100} = 0.5$.
- Use the formula: **Part = Percent $\times$ Whole**.



Worked Examples: Percents

Example 1: What is 25% of 200?

$$= 0.25 \times 200 = 50.$$

Example 2: 40 is what percent of 80?

$$\frac{40}{80} = 0.5 = 50\%.$$

Example 3: 20 is 25% of what number?

$$20 = 0.25x \Rightarrow x = 80.$$

Example 4: A 60 item is discounted by 15%. Find sale price.

$$0.15 \times 60 = 9. \text{ New price} = 60 - 9 = 51.$$

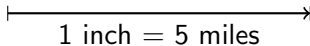
Example 5: A 1200 computer has 13% tax. Total cost?

$$0.13 \times 1200 = 156. \text{ Total} = 1356.$$

Lesson 4: Applications of Ratios, Proportions, and Percents

Objective: Apply ratios, proportions, and percents in real-world problems.

- Combine skills: compare quantities, solve for unknowns, and interpret percent changes.


$$1 \text{ inch} = 5 \text{ miles}$$

Worked Examples: Applications

Example 1: A map scale shows 1 inch = 5 miles. How many miles is 3.5 inches?

$$\frac{1}{5} = \frac{3.5}{x} \Rightarrow x = 17.5 \text{ miles.}$$

Example 2: A population grows from 800 to 1000. Find percent increase.
 $\frac{200}{800} = 0.25 = 25\%$.

Example 3: A student answered 42 of 50 questions correctly.
 $\frac{42}{50} = 0.84 = 84\%$.

Example 4: A recipe for 6 servings uses 3 cups of rice. How much for 10 servings?

$$\frac{6}{3} = \frac{10}{x} \Rightarrow x = 5 \text{ cups.}$$

Example 5: A car travels 240 miles on 8 gallons. How far on 10 gallons?
 $\frac{8}{240} = \frac{10}{x} \Rightarrow 8x = 2400 \Rightarrow x = 300 \text{ miles.}$

Review Summary

- Ratios compare quantities visually and numerically.
- Proportions show equivalent relationships.
- Percents express parts per hundred with circle or bar visuals.
- Apply all three concepts in real-world problems.